Sunkyu Shin

Postdoctoral Associate

Department of Chemical Engineering

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RESEARCH INTERESTS

"AI-Accelerated Multiscale Modeling of Petrochemical & Energy Systems"

[Quantum to Molecular Scale]

Detailed Kinetic Modeling for Clean Energy Systems

- Kinetic modeling for ammonia-based combustion systems.
- Automated kinetic mechanism generation using the Reaction Mechanism Generator (RMG).
- Integration of quantum chemistry results into detailed kinetic modeling.

[Equipment scale]

Multiphysics Reactor Modeling

- CFD-compartment modeling for industrial-scale reactors.
- Free raidcal polymerization modeling for molecular weight distribution (MWD).

[AI Integration]

Physics-Informed AI for Chemical Reactor Optimization

- AI-based surrogate model for efficient optimization of CFD models.
- Physics-informed neural networks (PINNs) for reactor flow prediction.
- Geometry-generalizable AI models for reactor design and optimization.

ACADEMIC APPOINTMENTS & EDUCATION

Aug. 2025– Present **Massachusetts Institute of Technology** Cambridge, MA, US

Postdoctoral Associate

Supervisor: Prof. William H. Green

Apr. 2024 – Jul. 2025 **Massachusetts Institute of Technology** Cambridge, MA, US

Postdoctoral Associate

Supervisor: Prof. Richard D. Braatz

Mar. 2020 – Feb. 2024 **Seoul National University** Seoul, South Korea

PhD in Chemical and Biological Engineering

Advisor: Prof. Won Bo Lee

Co-advisors: Prof. Myung-June Park and Prof. Jonggeol Na

Mar. 2018 – Feb. 2020 **Pohang University of Science and Technology (POSTECH)** Pohang, South Korea

Master of Engineering in Chemical Engineering

Advisor: Prof. In-Beum Lee

Mar. 2014 – Feb. 2018 **Pohang University of Science and Technology (POSTECH)** Pohang, South Korea

Bachelor of Engineering in Chemical Engineering, magna cum laude

PUBLICATIONS

*Corresponding author ; †These authors equally contributed.

1. **Sunkyu Shin**, Yu-Chi Kao, Timo T. Pekkanen, Yujie Qian, Yi-Pei Li, William H. Green*, "Kinetic Mechanism Development and Pollutant Chemistry in NH3/H2/CH4 Combustion", ***Journal of Physical Chemistry A***, 2026, 130 (40): 8011–8026, *https://doi.org/10.1021/acs.jpca.6c04337*.

2. Saikat Mukherjee*, **Sunkyu Shin**, Cedric Devos, Aniket Udepurkar, Peter Sagmeister, Bernhardt L. Trout, Allan S. Myerson, and Richard D. Braatz*, "Understanding Size Distributions during Lipid Nanoparticle Manufacturing through Mechanistic Modeling", *ACS Omega, 2026, 11, 26901–26908, https://doi.org/10.1021/acsomega.6c00100*.

3. Yu-Chi Kao, Anna C.Doner, Timo T.Pekkanen, Chuangchuang Cao, **Sunkyu Shin**, Alon Grinberg Dana, Yi-Pei Li, William H.Green*, "Detailed kinetic model for combustion of NH3/H2 blends", *Physical Chemistry Chemical Physics,* 2026, 28 (10): 6411–6424, *https://doi.org/10.1039/d5cp04149j*.

4. **Sunkyu Shin**†, Yubin Ryu†, Won Bo Lee*, and Jonggeol Na*, "Simultaneous optimization of reactor design and operating conditions using geometry-aware physics-informed neural network", ***Chemical Engineering Journal**, 2026, 527, 171870, https://doi.org/10.1016/j.cej.2025.171870*.

5. Mohammed Aatif Shahab, Nathan Merica Stover, Hasan Al-Mahayni, Prakitr Srisuma, Jie Wang, **Sunkyu Shin**, Anish Dighe, Bernhardt L. Trout, Allan S. Myerson, and Richard D. Braatz*, "End-to-end digital twin software for continuous mRNA manufacturing", *International Journal of Pharmaceutics*, 2025, 686, 126301, *https://doi.org/10.1016/j.ijpharm.2025.126301*.

6. **Sunkyu Shin**, Cedric Devos , Aniket Pradip Udepurkar , Pavan K. Inguva , Allan S. Myerson , Richard D. Braatz*, "Mechanistic modeling of lipid nanoparticle (LNP) precipitation via population balance equations (PBEs)", ***Chemical Engineering Journal***, 2025, 523, 167786, *https://doi.org/10.1016/j.cej.2025.167786*.

7. Pavan K. Inguva, Saikat Mukherjee, Pierre J Walker, Vico Tenberg, Cedric Devos, **Sunkyu Shin**, Yanchen Wu, Srimanta Santra, Jie Wang, Shalini Singh, Mona A Kanso, Shin Hyuk Kim, Bernhardt L Trout, Martin Z Bazant, Allan S Myerson, and Richard D Braatz*, "Mechanistic modeling of lipid nanoparticle formation for the delivery of nucleic acid therapeutics", *Biotechnology Advances*, 2025, 84, 108643, *https://doi.org/10.1016/j.biotechadv.2025.108643*.

8. **Sunkyu Shin**, Won Bo Lee*, and Myung-June Park*, "Surrogate-model-based optimization of a nonideal polymerization reactor for producing tailored molecular weight distribution", ***Chemical Engineering Journal**, 2024, 491, 152008, https://doi.org/10.1016/j.cej.2024.152008*.

9. Yubin Ryu†, **Sunkyu Shin**†, Won Bo Lee*, and Jonggeol Na*," Multiphysics generalization in a polymerization reactor using physics-informed neural networks", *Chemical Engineering Science*, *2024, 298, 120385, https://doi.org/10.1016/j.ces.2024.120385*.

10. **Sunkyu Shin**, Damin Lee, Ji Ye Oh, Youngjo Kim, Myung-June Park*, and Won Bo Lee*, "CFD modeling of the ethylene–vinyl acetate copolymerization autoclave reactor: Effects of comonomer ratio on reactor

dynamics and copolymer properties", *Chemical Engineering Science, 2023, 268, 118430, https://doi.org/10.1016/j.ces.2022.118430*.

11. Joonsoo Park, **Sunkyu Shin**, Youngjo Kim, Ji Ye Oh, and Jong Min Lee*, "Optimal Control of Polymer Grade Change in LDPE Autoclave Reactor based on Nonlinear Model Predictive Control", *The 22nd International Conference on Control, Automation and Systems (ICCAS 2022).*

12. **Sunkyu Shin**, Solji Choi, Jonggeol Na, Ikhwan Jung, Min-Kyu Kim, Myung-June Park*, and Won Bo Lee*, "CFD modeling for the prediction of molecular weight distribution in the LDPE autoclave reactor: Effects of non-ideal mixing", ***Chemical Engineering Journal**, 2022, 427, 131829, https://doi.org/10.1016/j.cej.2021.131829*.

13. Jeong-Keun Lee, **Sunkyu Shin**, Geun-Jae Kwak, Min-Kyung Lee, In-Beum Lee*, and Young-Seek Yoon, "Techno-economic evaluation of polygeneration system for olefins and power by using steel-mill off-gases", *Energy Conversion and Management, 2020, 224, 113316, https://doi.org/10.1016/j.enconman.2020.113316*.

14. **Sunkyu Shin**, Jeong-Keun Lee, and In-Beum Lee*, "Development and techno-economic study of methanol production from coke-oven gas blended with Linz Donawitz gas", ***Energy**, 2020, 200, 117506, https://doi.org/10.1016/j.energy.2020.117506*.

EXPERIENCE

Oct. 2018 – Oct. 2018 Imperial College London London, The United Kingdom
Carbon Capture Pilot Plant Operation and Control Program

Apr. 2017 – Aug.2017 Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen, Germany
Exchange Student Program

REGISTERED PATENTS

Won Bo Lee, **Sunkyu Shin**, Ikhwan Jung, and Myung-June Park, "Method for calculating polymer molecular weight distribution growth in non-ideal mixing", 2023, *Korea Patent registration*, *KR102941625B1*.

INVITED SEMINARS

Pohang University of Science and Technology (POSTECH), Pohang, South Korea, Feb 2022.

Pohang University of Science and Technology (POSTECH), Pohang, South Korea, Dec 2022.

HONORS & AWARDS

2014 - 2018 National Science & Technology Scholarship